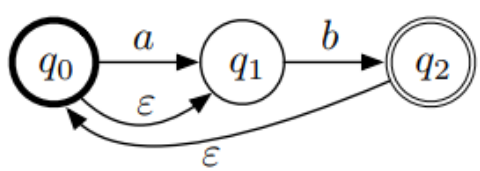
Assignment 1
NLP 201: Natural Language Processing I

All assignments are to be completed individually. You may discuss with the TAs and the instructor, but you may not receive help from anyone else.

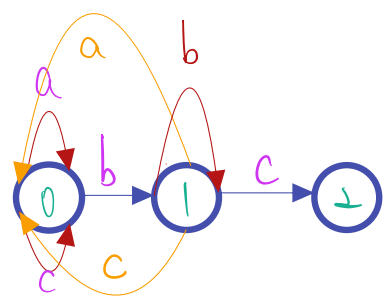
Instructions. Create a LaTeX-typeset write-up of your solutions and submit it to Canvas. Some free tools that might help: TexStudio (Windows), MacTex (Mac), TeX Live (cross-platform), and Overleaf (online). For the problems that require that you give FSAs, we encourage you to create them using drawing tools such as OmniGraffle (Mac) or yEd (online or cross-platform), but we will accept hand-drawn figures for full credit. Make sure initial and final states are clearly identified and explicitly list all symbols on each arc (i.e., no wildcards). We ask that you do not use any automated tools to *solve* these problems as you will not have access to such tools for similar problems on the exams.

Problem 1 [15 points]

For the following NFA containing ε-moves,



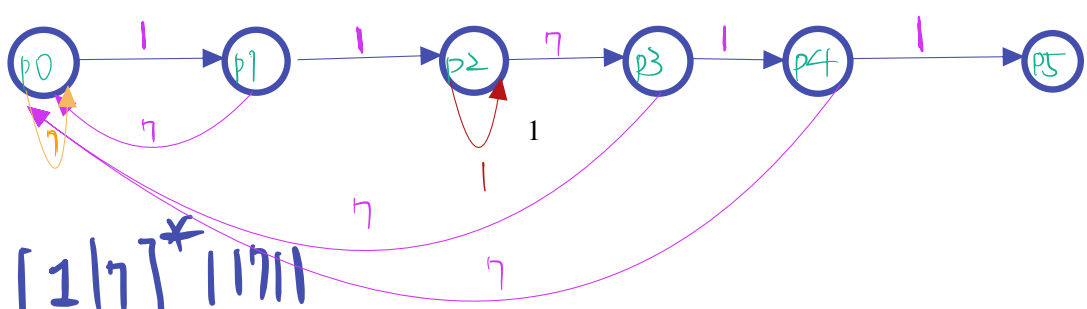
give an equivalent DFA.



Problem 2 [10 points]

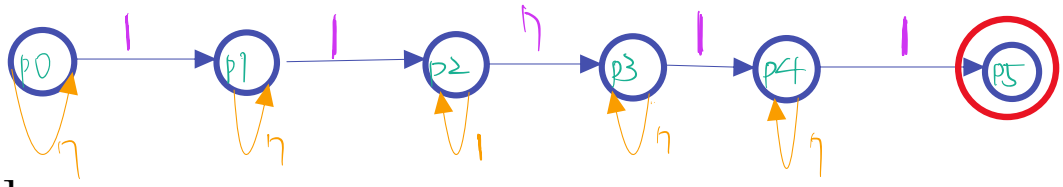
Give DFAs accepting the following languages.

1. **[5 points]** The set of strings over the alphabet $\{a, b, c\}$ in which the substring bc never occurs.
2. **[5 points]** The set of strings over the alphabet $\{1, 7\}$ that end in 11711.



11711

2. $[1|7]^*11711$



Problem 3 [10 points]

Give a *non-deterministic* FSA (possibly with ε-moves) accepting the following language.

1. The set of strings over $\{1, 7\}$ ending in 11711. How does your NFA differ from the DFA you constructed in Problem 2.2?

Problem 4 [15 points]

Give regular expressions for each of the following languages:

1. **[5 points]** The set of strings over $\{1, 7\}$ ending in 711. $(1|7)^* 711$
2. **[5 points]** The set of strings over $\{a, b, c\}$ containing the substring ab or bc. $(a|b|c)^* ab|bc (a|b|c)^*$
3. **[5 points]** The set of strings over $\{a, b, c\}$ either starting with a and containing the substring ab *or* starting with b and containing the substring bc.

$(a)^* ab | (b)^* bc$

Problem 5 [20 points]

We may define **generalized regular expressions** (GREs) as follows:

1. $\emptyset$ is a GRE denoting the empty language;
2. ε is a GRE denoting the language $\{\varepsilon\}$;
3. for each $\sigma \in \Sigma$, σ is a GRE denoting the language $\{\sigma\}$;
4. if α and β are GREs, denoting the languages A and B, respectively, then
   - $(\alpha \mid \beta)$ is a GRE denoting $A \cup B$;
   - $(\alpha\beta)$ is a GRE denoting $A.B$;
   - α^* is a GRE denoting A^*;
   - **[new]** $(\alpha \wedge \beta)$ is a GRE denoting $A \cap B$; and
   - **[new]** $\neg\alpha$ is a GRE denoting $\overline{A}$.

Prove that the languages denoted by GREs are regular.

Note: you may use refer to the definitions and proofs about REs we provided in class and only deal with the two new clauses in the generalized definition.

Problem 6 [25 points]

For each of the following statements, answer whether the claim is **true** or **false**, and give a *short* (one- to two-sentence) explanation if true or counterexample if false.

1. **[5 points]** Let $L_1 \subset L_2$. If L_1 is not regular, then L_2 must also be not regular.

2. **[5 points]** $L = L_1 \cap L_2$. If L_1 and L are regular languages, then L_2 must also be a regular language.

3. **[5 points]** $L = L_1 \cup L_2$. If L_1 and L are regular languages, then L_2 must also be a regular language.

4. **[5 points]** $L = \bigcap_{i=1}^{\infty} L_i$. If all of the L_i are regular languages, then L is also a regular language.

5. **[5 points]** Let α, β and γ be regular expressions.
   If $L(\beta \mid \alpha\gamma) \subseteq L(\gamma)$, then $L(\alpha^*\beta) \subseteq L(\gamma)$.

5: RL = ① $L \subseteq \Sigma^*$

② have FSA (M) accept $L(M) = L$

如果 FSA 可以接受 L 那 L 就是 RL

* $L = \Sigma^*$ Σ = 字母表上的所有字串.

$L = \emptyset$ $\emptyset$ = 空語言

$L = \{\varepsilon\}$ $\{\varepsilon\}$ = 空字串.

RE = 描述 FSA 接受語言的表達式.

RE ① $\emptyset$ FSA accept $\emptyset$

② $\{\varepsilon\}$ FSA accept $\{\varepsilon\}$

③ Σ^* FSA accepe each [words or char] from Σ

④ $(\alpha|\beta)$

RE is closed union (封閉的), so 2 RE still is RE

⑤ $(\alpha\beta)$

RE is closed union, so 2 RE still RE

⑥ $(\alpha)^*$

RE is closed union, so n time of RE still RE

⑦ A is RE B is RE.

A Have FSA call M(A)

B Have FSA call M(B)

def M(AB) is cartesian product, states (q_A, q_B) = q_A is in M(A) q_B is in M_B

Init M(AB) is pair of Init M(A) and Init M(B)

Accept state is pair All Q(A) and Q(B)

The $M(AB)$ have $\delta_{(AB)}$ is from (q_A, q_B) to (q_A', q_B')

And (q_A') is transition in $M(A)$

(q_B') is transition in $M(B)$

所以我们可以定义 $(\alpha \wedge \beta)$ (A与B的交集) $A \cap B$)

8 $-A$ is FSA.

We know A is RE and have $M(A)$

① build a $M(-A)$, we need to chang All $M(A)$ accepting to Accepting state

than we build $M(-A)$ $\Rightarrow$ $M(-A)$ accept All the string that $M(A)$ not accepting

So $-A$ also the closed union.

6

① F

$L_1 = \{a^n b^n \mid n \geq 0\}$ not RL

$L_2 = a^n b^n$ is RL.

$L_1 \subset L_2$ L_2 is RL L_1 not RL.

② $L = L_1 \cap L_2$ is F.

$L_1 = a^*$ is RL

$L_2 = \{a^n b^n \mid n \geq 0\}$ not RL

$L = L_1 \cap L_2 = \{a^n \mid n \geq 0\}$, but not RL

③ $L = RL = L_1 \cup L_2$ L_1 is RL L_2 need to? F

$L_1 = a^*$

$L_2 = \{a^n b^n \mid n \geq 0\}$ not RL

$L = L_1 \cap L_2 = a^* \cup \{a^n b^n \mid n \geq 0\}$

$\Rightarrow L = a^* b^*$ is RL

④ $L = \bigcap_{i=1}^{\infty} L_i$

$L_i = a^* b^* c^* = RL$

L_i Automaton $\Rightarrow L = \bigcap_{i=1}^{\infty} L_i$

than just include some combination

So is F.

⑤ α, β, γ is RL

If $L(\beta | \alpha\gamma) \subseteq L(\gamma)$

$L(\alpha^* \beta) \subseteq L(\gamma)$ is F

If $\alpha = a$ $\beta = b$ $\gamma = ab$.

then $L(\beta | \alpha\gamma) = L(b | aab) \subseteq L(ab)$

but $L(\alpha^* \beta) = L(a^* b)$

a include several combination

If γ from γ string